

1 KNN Algorithm Implementation

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[1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import math
from collections import Counter

data = pd.read_csv("Data.csv")
data
```

```
[1]:
```

	Observation	X1	X2	X3	Y
0	1	0	3	0	Red
1	2	2	0	0	Red
2	3	0	1	3	Red
3	4	0	1	2	Green
4	5	-1	0	1	Green
5	6	1	1	1	Red

```
[2]: # Step 2: Define Euclidean distance function
def euclidean_distance(p1, p2):
    return math.sqrt(sum((a - b) ** 2 for a, b in zip(p1, p2)))
```

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[3]: #Step 3: Define KNN function
def knn_classify(train_data, y_train, test_point, k):
    distances = []
    # for row in train_data:
    #     features = [row["X1"], row["X2"], row["X3"]]
    #     dist = euclidean_distance(features, test_point)
    #     distances.append((dist, row["Y"]))

    for i in range(len(train_data)):
        dist = euclidean_distance(test_point, train_data[i])
        distances.append((dist, y_train[i]))

    print(distances)
    # Step 4: Sort distances and pick k nearest
    distances.sort(key=lambda x : x[0])
    k_nearest = [label for _, label in distances[:k]]

    # Step 5: Majority vote
    most_common = Counter(k_nearest).most_common(1)
    return most_common[0][0]
```

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[4]: train_data = data.iloc[:, 1:-1].values
y_train = data.iloc[:, -1].values

# Step 6: Example test point
test_point = [0, 0, 0]

print(train_data[0])
print(test_point)
k = 3
```

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predicted_label = knn_classify(train_data, y_train, test_point, k)
```

```
[0 3 0]
```

```
[0, 0, 0]
```

```
[(3.0, 'Red'), (2.0, 'Red'), (3.1622776601683795, 'Red'), (2.23606797749979,  
'Green'), (1.4142135623730951, 'Green'), (1.7320508075688772, 'Red')]
```

```
[5]: predicted_label
```

```
[5]: 'Red'
```